

Lecture 6: Dynamic Programming for Graphs

usually the trickiest part

The DP Recipe

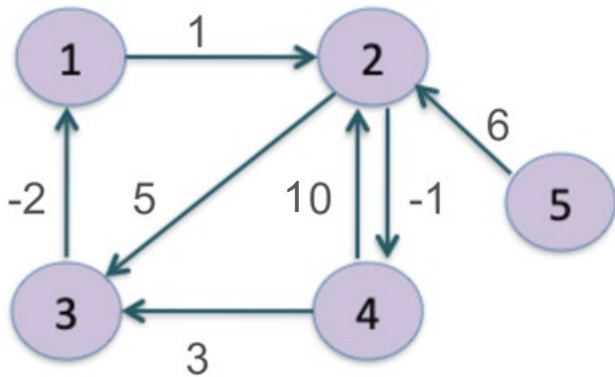
1. Define *value* version of what we want to compute
2. Derive a recurrence
3. Size of table: How many dimensions? Range of each dimension?
4. What are the base case(s)?
5. To fill in a cell, which other cells need to be filled already? In which order do I fill the table?
6. Which cell(s) contain the final answer?
7. Running time is often (size of table) $\cdot$ (time to fill each entry)
8. To reconstruct the solution (instead of just its value) follow arrows from final answer to base case

The shortest-path problems we'll consider

Input: Weighted directed graph. Weights can be negative, but _____

Single-Source Shortest Paths (SSSP): Given a “source” vertex s , find a shortest path from s to every vertex t .

All-Pairs Shortest Paths (APSP): For every pair s, t of vertices, find a shortest path from s to t .



	1	2	3	4	5
1	0	1	3	0	∞
2	0	0	2	-1	∞
3	-2	-1	0	-2	∞
4	1	2	3	0	∞
5	6	6	8	5	0

What about single-pair shortest path?



Recurrence for SSSP

Key Definition. Let $\text{dist}^{(i)}(s, v)$ be the *length* of a shortest s-to-v path that uses at most i edges, or ∞ if there's no such path.

Ultimate goal: $\text{dist}^{(n-1)}(s, v)$ for each v . Why?

Recurrence:

$$\text{dist}^{(i)}(s, v) =$$

Base case(s):

The recurrence will be similar in structure to Professor Y's first attempt



This yields the **Bellman-Ford Algorithm**
(Bellman, Ford, Moore, Shimbel, 1955)

Old idea: “ i ” parameter counts length of permissible paths.

New idea: Order vertices arbitrarily as $v_1, v_2, \dots, v_n$. New “ i ” parameter says *allowable intermediate vertices* are $v_1, \dots, v_i$.

Key Definition:

$dist^{[i]}(s, t)$ = length of shortest s - t path that is allowed to use vertices $\{s, t\} \cup \{v_1, \dots, v_i\}$ (not necessarily all of them).

$$dist^{[i]}(s, t) = \min \left\{ \begin{array}{l} dist^{[i-1]}(s, t) \\ \min_{v_j \in \{v_1, \dots, v_i\}} \{ dist^{[i-1]}(s, v_j) + dist^{[i-1]}(v_j, t) \} \end{array} \right\}$$

Use it!

Or lose it!

Base case: $dist^{[0]}(s, t) =$